EE1

Friday, 9 October 2026 8:00 am

Scriviamo la matrice

$$A = \begin{pmatrix} 0 & 2 & -3 \\ \textcircled{1} & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 3 & 2 \end{pmatrix}$$

come somma di r matrici di rango 1, dove $r = \text{rank}(A)$.

$$A = \begin{pmatrix} \textcircled{1} & 1 & 1 \\ 0 & 2 & -3 \\ 2 & 3 & 1 \\ 1 & 3 & 2 \end{pmatrix} \mapsto \begin{pmatrix} \textcircled{1} & 1 & 1 \\ 0 & 2 & -3 \\ 0 & 1 & -1 \\ 0 & 2 & 1 \end{pmatrix} \mapsto \begin{pmatrix} \textcircled{1} & 1 & 1 \\ 0 & \textcircled{1} & -1 \\ 0 & 2 & -3 \\ 0 & 2 & 1 \end{pmatrix} \mapsto \begin{pmatrix} \textcircled{1} & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & -1 \\ 0 & 0 & -1 \end{pmatrix}$$

$$\mapsto \begin{pmatrix} \textcircled{1} & 1 & 1 \\ 0 & \textcircled{1} & -1 \\ 0 & 0 & \textcircled{-1} \\ 0 & 0 & 0 \end{pmatrix} = U$$

$$A = S_{12} L_{31}(2) L_{41}(1) S_{23} L_{32}(2) L_{42}(2) L_{43}(-1) \quad U$$

$$= S_{12} L_{31}(2) S_{23} L_{41}(1) L_{32}(2) L_{42}(2) L_{43}(-1) U$$

$$= S_{12} S_{23} L_{21}(2) L_{41}(1) \dots \quad U$$

$$= \begin{pmatrix} e_3^T \\ e_1^T \\ e_2^T \\ e_4^T \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 1 & 2 & -1 & 1 \end{pmatrix} U$$

$$= \begin{pmatrix} 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 1 & 2 & -1 & 1 \end{pmatrix} \begin{pmatrix} \textcircled{1} & 1 & 1 \\ 0 & \textcircled{1} & -1 \\ 0 & 0 & \textcircled{-1} \\ 0 & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & 1 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 2 & -2 \\ 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 2 & -2 \end{pmatrix} + \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & +1 \end{pmatrix}$$